SOFTWARE REQUIREMENTS SPECIFICATION · VERSION 1.0

Pothgula

A web-based lending and membership system for a small library operating across several branches.

PREPARED BY	ROLE	DATE	STATUS
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01

Purpose and scope

Pothgula replaces the paper register. It keeps one shared record of every book, every member and every loan across all of your branches, and it tells your staff what needs attention today.

What the system does

- Holds the **catalogue** of every book copy the library owns, with the branch that holds it and whether it is on the shelf.
- Records **who borrowed what and when it is due back**, and shows overdue loans to staff.
- Tracks **membership** — each member is valid for one year and is renewed when the annual fee is paid.
- Publishes each branch's **opening days and times** to members.
- Warns members before a book is due and before their membership runs out.

What it does not do

This release is a lending and membership system, not a full library automation package. It does not do acquisitions and purchase orders, inter-library loans, digital or e-book lending, online fee payment, or a public catalogue open to non-members. Section 13 explains the consequences of each of these; section 14 shows where they could be added later.

> **HOW TO READ THIS DOCUMENT**
>
> Every requirement has an identifier such as `FR-CIR-04` and a priority. **MUST** means the system is not accepted without it. SHOULD means it is planned and will be dropped only if we agree to drop it. COULD means it is desirable and will be built if time allows. Please mark any requirement you disagree with before signing section 16.

02

Who uses the system

Three kinds of people use Pothgula. Everyone signs in with their phone number and a password.

MEMBER

The person who borrows books

Registers once, pays an annual fee, and belongs to one home branch.

- Browses the catalogue and sees what is on the shelf
- Checks the books they currently hold and when they are due
- Sees opening hours and their own membership expiry

ADMIN · LIBRARIAN

Runs the counter at one branch

Day-to-day staff. Everything they do is limited to their own branch.

- Issues books and takes them back
- Records annual fee payments and renews memberships
- Sets that branch's opening days and times

SUPER ADMIN

Runs the library organisation

Everything an admin can do, plus the setup that admins must not touch.

- Creates branches and staff accounts
- Adds and edits books in the catalogue
- Works across every branch, not just one

Who can do what

CAPABILITY	MEMBER	ADMIN	SUPER ADMIN
Browse the catalogue	**Yes**	**Yes**	**Yes**
See other branches' books	**Yes**	**Yes**	**Yes**
Issue a book to a member	—	**Own branch**	**Any branch**
Take a book back	—	**Own branch**	**Any branch**
Correct a loan record	—	**Own branch**	**Any branch**
Record an annual fee payment	—	**Own branch**	**Any branch**
Set opening days and times	—	**Own branch**	**Any branch**
Add or edit a book	—	—	**Yes**
Create a member account	Self only	—	**Yes**
Create a staff account	—	—	**Yes**
Create or edit a branch	—	—	**Yes**

Admins cannot add books in this design, only super admins can. If your librarians are expected to catalogue new arrivals themselves, tell us — this is question Q3 in section 15.

03

Accounts and access

ID	REQUIREMENT	PRIORITY
FR-AUT-01	The landing page lets a visitor choose Member, Admin or Super admin, then choose to sign in or register.	MUST
FR-AUT-02	Sign-in uses phone number and password. Passwords are stored one-way hashed and can never be read back by anyone, including us.	MUST
FR-AUT-03	A member registers with their name, phone number, address and home branch. Their membership is dated one year ahead.	MUST
FR-AUT-04	Registering as a super admin requires the pre-shared super admin code. At most five super admin accounts may exist per branch.	MUST
FR-AUT-05	A super admin can create an account of any type directly, without the new person registering themselves.	MUST
FR-AUT-06	An account is locked for 15 minutes after five failed sign-in attempts. The same limit applies to attempts at the super admin code.	SHOULD
FR-AUT-07	A staff session ends after 30 minutes of inactivity, so a counter machine left unattended does not stay signed in.	SHOULD
FR-AUT-08	A member can reset a forgotten password using a code sent to their registered phone number.	SHOULD
FR-AUT-09	A super admin can deactivate a staff or member account. Accounts are never deleted, so their loan history stays intact.	SHOULD

> PLEASE NOTE
>
> Requirement FR-AUT-04 means the super admin code is the single thing standing between the internet and full control of your library data. Recommendation R5 in section 12 proposes a safer arrangement at no extra cost.

04

The book catalogue

Every physical book on your shelves is one entry in the catalogue, identified by a number written or stuck on the book itself.

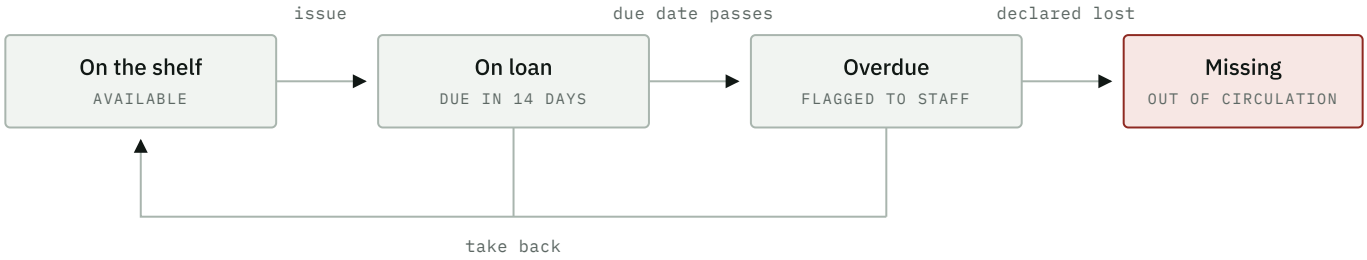
ID	REQUIREMENT	PRIORITY
FR-CAT-01	A super admin registers a book with its title, accession number, category letter, writer, publisher, and how the library obtained it (purchase, donation, or the donor's name).	MUST
FR-CAT-02	The accession number is one capital letter followed by five digits, for example `A12345`. The system rejects any other format and refuses a number already in use anywhere in the library.	MUST
FR-CAT-03	Each book belongs to exactly one branch, and its condition is recorded as Useable, Missing or Not useable.	MUST
FR-CAT-04	Each book shows as On the shelf or On loan. A book that is missing or not useable can never be issued.	MUST
FR-CAT-05	A member sees their home branch's catalogue by default and can filter it by category, writer and availability.	MUST
FR-CAT-06	A member can switch the view to any other branch and see which branch holds a copy of a book.	MUST
FR-CAT-07	A search box finds books by title, writer or accession number, across one branch or all branches.	MUST
FR-CAT-08	A super admin can edit any detail of a book, including moving it to a different branch.	MUST
FR-CAT-09	Category letters A to Z can each be given a readable name, so members see "Sinhala fiction" rather than "F".	SHOULD
FR-CAT-10	The system can print a spine label carrying the accession number as a scannable barcode.	COULD

> **ADDITION TO YOUR ORIGINAL DESIGN**
>
> FR-CAT-01 adds a **book title**, which the original table list did not include. A catalogue that a member can search by writer and category but not by title is not usable at the counter, so we have treated the title as essential rather than optional. Recommendation R1 goes further and explains why the title should be separated from the physical copy.

05

Issuing and taking back

This is the part of the system your staff will use every day, so it is designed around the counter: find the member, find the book, confirm.



The life of one book copy. Every arrow is an action a staff member takes or a date passing; nothing changes state on its own except the move to Overdue.

ID	REQUIREMENT	PRIORITY
FR-CIR-01	Staff issue a book by selecting a member and a book. The system records today as the borrow date and sets the due date 14 days later.	MUST
FR-CIR-02	The system refuses to issue a book and states the reason if it is already on loan, missing, not useable, the member's membership has expired, or the member is already holding the maximum number of books.	MUST
FR-CIR-03	Staff take a book back in one action. The return date is recorded and the book returns to the shelf immediately.	MUST
FR-CIR-04	If a book comes back after its due date, the system shows how many days late it was and keeps that on the loan record.	MUST
FR-CIR-05	Staff can correct a loan record — wrong book, wrong member, wrong date. Every correction records who made it and when.	MUST
FR-CIR-06	The maximum number of books one member may hold at a time is a setting, defaulting to two.	MUST
FR-CIR-07	The loan period is a setting per branch, defaulting to 14 days, so a branch can run a longer or shorter period without new software.	SHOULD
FR-CIR-08	At take-back, staff can mark a book as damaged or lost instead of returning it to the shelf. Its condition changes and the loan stays open against the member until it is settled.	SHOULD
FR-CIR-09	A member's page shows their complete borrowing history, most recent first.	SHOULD
FR-CIR-10	A loan can be renewed once for a further loan period, provided it is not already overdue.	COULD

06

Membership and annual fee

ID	REQUIREMENT	PRIORITY
FR-MEM-01	A membership is valid for one year. The first year runs from the date the member registers.	MUST

ID	REQUIREMENT	PRIORITY
FR-MEM-02	Staff record an annual fee payment against a member. The membership is extended by one year from its previous expiry date, so a member who pays late does not gain extra time.	MUST
FR-MEM-03	Staff see three lists on one screen: paid in the last 30 days, expiring in the next 30 days, and already expired.	MUST
FR-MEM-04	An expired member can still sign in and browse the catalogue but cannot be issued a book until the fee is paid.	MUST
FR-MEM-05	Each payment is kept as its own record with the amount, the date, a receipt number and the staff member who collected it. Payments are never overwritten.	SHOULD
FR-MEM-06	The annual fee amount is a setting per branch rather than a fixed number in the software.	SHOULD
FR-MEM-07	The system can print or export a receipt for a payment.	COULD

07

Opening hours

ID	REQUIREMENT	PRIORITY
FR-SCH-01	Staff set a weekly pattern for their branch: for each day of the week, an opening and closing time, or closed.	MUST
FR-SCH-02	Staff can mark specific dates as closed with a reason, for Poya days, public holidays and stock-taking. These override the weekly pattern.	MUST
FR-SCH-03	A member sees their home branch's hours on their dashboard and can look up any other branch's hours.	MUST
FR-SCH-04	Today's status is stated plainly: Open until 5.00 pm, Opens at 9.00 am, or Closed today, with the reason where one is recorded.	SHOULD
FR-SCH-05	A closure can be set for a date range in one action rather than day by day.	COULD

08

Alerts and reminders

ID	REQUIREMENT	PRIORITY
FR-NOT-01	A member's dashboard warns them three days before a book is due, and shows a stronger warning every day it stays overdue.	MUST
FR-NOT-02	A member is warned 30 days before their membership expires, and told clearly once it has expired.	MUST
FR-NOT-03	The staff dashboard opens on today's work: books due back today, books already overdue, and memberships expiring this month.	MUST

ID	REQUIREMENT	PRIORITY
FR-NOT-04	Staff can produce a call list of overdue members with their phone numbers, for following up by telephone.	SHOULD
FR-NOT-05	The system sends an SMS reminder for a due book and an expiring membership.	COULD

> **COST NOTE**
>
> FR-NOT-05 needs a paid SMS gateway account, billed per message by a Sri Lankan provider. That is an ongoing cost you would carry directly, not part of the software price. Until it is enabled, a member only sees a warning when they sign in — which the members you most need to reach are the least likely to do.

09

Reports

ID	REQUIREMENT	PRIORITY
FR-RPT-01	Stock summary for a branch: how many books in each category, and how many are useable, missing or not useable.	SHOULD
FR-RPT-02	Currently issued and overdue lists, downloadable as a spreadsheet file.	SHOULD
FR-RPT-03	Fee collection summary for a chosen date range, per branch and in total.	SHOULD
FR-RPT-04	Membership count over time: new members, renewals and lapsed members per month.	SHOULD
FR-RPT-05	Most borrowed books and least borrowed books over a chosen period, to guide what you buy next.	COULD

10

Quality requirements

These apply to the whole system rather than to one feature, and they are as much a part of acceptance as the features above.

ID	AREA	REQUIREMENT
NFR-01	Speed	Any counter action — search, issue, take back — completes in under two seconds with 50 people using the system at once.
NFR-02	Devices	Works on a desktop browser at 1366×768 and on a phone. Members will mostly use phones; staff will mostly use the counter machine.

ID	AREA	REQUIREMENT
NFR-03	Browsers	Current versions of Chrome, Edge, Firefox and Safari.
NFR-04	Language	Sinhala, Tamil and English text can be entered and displayed correctly in every field, including names and book titles.
NFR-05	Time	All dates and times are shown in Sri Lanka time. Due dates are calculated by date, not by hour, so a book issued at 4.55 pm is due on the same day of the fortnight as one issued at 9.05 am.
NFR-06	Security	All traffic over HTTPS. Passwords hashed with a modern algorithm. Staff pages unreachable without a valid staff session, checked on the server for every request.
NFR-07	Audit	Every staff action that changes data records who did it and when, and that record cannot be edited from within the application.
NFR-08	Backup	The database is backed up daily and a restore is tested before handover. Recovery point: at most 24 hours of data.
NFR-09	Privacy	Member names, phone numbers and addresses are personal data under Sri Lanka's Personal Data Protection Act No. 9 of 2022. Access is limited to staff of the member's own branch and to super admins.
NFR-10	Correctness	The same book cannot be issued twice at the same moment, even if two staff members act at once. The system enforces this in the database, not only in the screen.

11

Information we store

This is the record structure behind the system. It is included so you can confirm nothing you need is missing before we build. Fields shown in green are additions to your original list that this document proposes.

branch

id	number
library_name	text
branch_name	text
address	text
phone	text
created_at / updated_at	date

book

id	A12345
title	text
category	A–Z
writer	text, optional
publisher	text, optional
source	text, optional
status	useable / missing / not useable
availability	on shelf / on loan
branch_id	branch
created_at / updated_at	date

user

id	number
full_name	text
user_type	member / admin / super admin
phone	text, unique
password	hashed
address	text
branch_id	branch
expire_at	date
is_active	yes / no
created_at / updated_at	date

borrow

id	number
book_id	book
user_id	member
branch_id	branch
borrow_date	date
due_date	borrow + 14 days
returned_at	date, empty while on loan
issued_by	staff user
received_by	staff user
created_at / updated_at	date

membership_payment

id	number
user_id	member
amount	LKR
paid_on	date
receipt_no	text
collected_by	staff user
extends_to	new expiry

branch_schedule

branch_id	branch
weekday	Mon–Sun
open_time	time, empty = closed
close_time	time

branch_closure

branch_id	branch
closed_on	date
reason	text

audit_log

actor_id	staff user
action	issued / returned / edited
target	record affected
before / after	values changed
at	date and time

Four of these tables — payments, opening hours, closures and the audit log — do not exist in the original design but are required by features you asked for. Without `branch_schedule` there is nowhere to put opening times; without `membership_payment` a renewal overwrites the previous expiry date and the payment history is lost.

What we recommend changing

These are our professional recommendations on the design as it stands. None of them is a defect in your thinking — they are the places where a decision that is reasonable today becomes expensive in two years. Each shows the effort to do it now.

ID	RECOMMENDATION	WHY IT MATTERS	EFFORT NOW
R1	Separate the **title** from the **copy**.	Today, five copies of the same book mean typing the writer, publisher and title five times, and a typo in one of them splits the book into two entries in search results. Storing the title once and the five accession numbers against it fixes this and makes "3 of 5 copies available" possible.	Low, before build
R2	Work out availability from the loans rather than storing it as a separate yes/no field.	Two places recording the same fact will eventually disagree — a crash mid-transaction leaves a book marked on loan with no loan against it, and staff cannot issue a book that is physically in their hand. One source of truth removes the whole class of problem.	Low, before build
R3	Store phone numbers as **text**, not as numbers, and require them to be unique.	Stored as a number, `0771234567` loses its leading zero and becomes `771234567`. Since the phone number is also the sign-in name, duplicates must be impossible.	Low, before build
R4	Drop the Member / Admin / Super admin choice from the **sign-in** screen; keep it only for registration.	The password already determines who someone is. Asking first only creates a way to fail — a librarian who picks Member gets told their password is wrong. It also confirms to a stranger that a given phone number belongs to an admin.	Low, before build
R5	Make the super admin code **single-use**. After the first super admin exists, further staff are created by an existing super admin, not by self-registration.	A fixed code shared among staff spreads by WhatsApp and never expires. Anyone who ever sees it can create a super admin account for themselves, at any time, from anywhere. A single-use bootstrap code closes the door permanently after setup.	Low, before build
R6	Have staff **approve** a member's self-registration at the counter before the membership becomes active, and start the year from that approval.	Otherwise anyone on the internet can create an account, and you cannot tell a real member from a name typed into a form. Approving at the counter is also the natural moment to check identity and collect the first fee.	Low, before build
R7	Record a **due date** and a separate **returned date** on each loan.	The original design has one date field whose meaning changes depending on whether the book is back. That makes "which books are overdue right now" impossible to answer reliably, which is the single most important question the system exists to answer.	Low, before build
R8	Decide the **rule for other branches**: can a member borrow a book held elsewhere, or only see it?	The design lets members search other branches but never says what happens next. If they can only look, members will travel to a branch expecting a book; if they can borrow, we need transfer requests and a way to track a book in transit. Either is fine, but it must be decided before build.	Low to decide, medium to build
R9	Make super admin an **organisation-wide** role, and admin the branch-scoped one.	As written, super admins are capped at five per branch yet their job is to create branches — so the first one has no branch to belong to. Organisation-wide super admins with branch-scoped librarians matches how the library actually works, and three or four super admins in total is safer than five per branch.	Low, before build

ID	RECOMMENDATION	WHY IT MATTERS	EFFORT NOW
R10	Add **barcode scanning** at the counter once the basics are live.	The accession number format is already ideal for a barcode. A USB scanner costs little, and issuing a book becomes scan-scan-done instead of two searches. This is the single biggest improvement to daily staff experience.	Medium, phase 3

13

Limitations and risks

What this system will not do for you, and what could go wrong. We would rather you know now than discover it after handover.

Serious — please decide on these before we build

Anyone with the code becomes an owner D1

The super admin registration code is a single shared secret with no expiry. A staff member who leaves keeps the ability to create a super admin account and read every member's phone number and address.

Mitigation: adopt R5. Otherwise, rotate the code whenever anyone leaves and accept the risk in writing.

Unverified members can borrow real books D2

Self-registration creates an active member instantly. There is no check that the person exists, that the address is real, or that the fee was paid — but the books they take away are physical and unrecoverable.

Mitigation: adopt R6, so registration online only produces an application that a librarian activates in person.

Overdue books have no consequence D3

The system will tell you a book is 40 days late, and that is all it does. There is no fine calculation, no borrowing block after a threshold, and no record of what was eventually recovered.

Mitigation: tell us your fine policy and we will cost it. At minimum, blocking further borrowing while a book is overdue costs almost nothing to add.

The counter stops when the internet does D4

This is a web system with no offline mode. During an outage at a branch, staff cannot issue or take back books and must record transactions on paper, then enter them later — which is when mistakes happen.

Mitigation: a printed daily loans sheet as a fallback, plus an agreed procedure for entering a backlog. Offline-capable software is a much larger project.

Worth knowing

No reservations or holds D5

A member who wants a book that is on loan has no way to queue for it. They must keep checking the catalogue, and whoever happens to arrive first gets it.

Option: phase 3 feature, moderate effort.

Reminders only reach people who sign in D6

Due-date and expiry warnings appear on the member's dashboard. A member who never signs in never sees them, and those are exactly the members who go overdue.

Option: SMS reminders, FR-NOT-05, with a per-message cost you carry.

Books cannot move between branches D7

A super admin can reassign a book to another branch, but there is no request, no approval and no in-transit state. A book being carried between branches simply appears to be on the shelf at its destination.

Option: depends on the answer to R8.

Personal data brings legal duties D8

Holding names, phone numbers and addresses makes the library a data controller under the Personal Data Protection Act No. 9 of 2022. You will need a stated purpose, a retention period and a way to respond if a member asks for their data.

Mitigation: we will supply a privacy notice for the registration page; the policy decisions are yours.

The catalogue is local, not standard D9

Books are catalogued with your own letter categories rather than ISBN and a standard classification. That is simpler to use, but it means the data cannot be imported into a standard library system later without manual work.

Option: add an optional ISBN field now — cheap insurance, no extra work at the counter.

Capacity of the numbering scheme D10

The `A12345` format allows 100,000 numbers per letter. That is ample, but the letter in the number is not the same as the category letter, and staff will assume it is. Agree now what the letter means.

Mitigation: either make the letter the category, or document that it is meaningless. Do not leave it ambiguous.

14

Delivery in three phases

We recommend putting a working counter system in your hands early rather than delivering everything at once. Each phase is usable on its own.

1

The counter works

FIRST RELEASE

Branches, accounts and sign-in, the book catalogue with search, issuing and taking back with due dates, membership expiry, and the staff dashboard showing today's overdue books. At the end of this phase you can stop using the paper register.

2

The library runs itself

SECOND RELEASE

Opening hours and closures, annual fee payments with history and receipts, member warnings for due dates and expiry, the member-facing dashboard, and the reports in section 9.

3

Faster and further

ONCE PHASES 1 AND 2 ARE SETTLED IN USE

Barcode scanning at the counter, SMS reminders, reservations, fines, renewals, and inter-branch requests. We would rather choose from this list after you have used the system for a few months than guess now.

15

Decisions we need from you

Each of these changes what we build. Answers to Q1 to Q4 are needed before development starts.

Q1	Can a member borrow a book held at another branch, or only see that it exists there? (Recommendation R8.)
Q2	Should a member's registration become active immediately, or only after a librarian approves it at the counter? (Recommendation R6.)
Q3	Should librarians be able to add and edit books, or is that strictly a super admin task? Cataloguing new arrivals is usually a librarian's job.
Q4	Is there a fine for overdue books? If so, what is the daily rate and is there a maximum?
Q5	How many books may one member hold at once? We have assumed two.

Q6	What is the annual membership fee, and is it the same at every branch?
Q7	How many branches will there be at launch, and roughly how many books and members in total?
Q8	Who will be the first super admin, and who holds the setup code?
Q9	Do you want SMS reminders in scope, given the per-message cost? (FR-NOT-05.)
Q10	Should the interface be in English only, or English with Sinhala and Tamil?

16

Acceptance

Signing below confirms that the requirements marked MUST in sections 3 to 10 describe the system you want built, that the limitations in section 13 have been read and accepted, and that the answers to section 15 have been provided in writing. Anything not listed in this document is outside the agreed scope and will be quoted separately.

Client — name, signature and date

Chathuranga Jayawardhana
System Designer — signature and date

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Questions on any part of this document are welcome before signing. It is easier to change a requirement now than a screen later.